Repertoire diversity and concentration: results

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Human is compared with GPT and with Claude on how varied the set of syntactic building blocks is and how concentrated its use is. Table 1 is the main result; Tables 2 and 3 are checks on it.

Table 1. Main result: each source at the same amount of text

This is the headline comparison. Human, GPT and Claude are each compared at the same amount of text (413,819 tokens in Student essays, 358,204 tokens in Reuters news and 368,665 tokens in WritingPrompts), so that no source is credited with more variety just because more of its text was read. The building blocks are the smallest kind: a phrase and its immediate parts, such as `(NP (DT) (NN))`, a noun phrase made of a determiner and a noun. Documents were drawn at random, whole, 1,000 times per source, and each value is the average over the draws. Reuters draws are by whole author.

Domain	Measure	Human	GPT	Claude
Student essays	Distinct blocks used (`types`)	6,954 ± 305	4,509 ± 176 (−35%)	6,233 ± 160 (−10%)
	Blocks used 5+ times (`types_min5`)	1,389 ± 42	1,148 ± 36 (−17%)	1,367 ± 43† (−2%)
	Effective number of blocks (`effective_types`)	133 ± 4	113 ± 2 (−15%)	134 ± 2† (+1%)
	Share taken by the 50 commonest (`cover_50`)	70.5% ± 0.5	73.1% ± 0.3 (+2.6 pts)	70.0% ± 0.3† (−0.5 pts)
Reuters news	Distinct blocks used (`types`)	7,190 ± 303	4,213 ± 171 (−41%)	6,002 ± 164 (−17%)
	Blocks used 5+ times (`types_min5`)	1,572 ± 53	1,211 ± 40 (−23%)	1,482 ± 45 (−6%)

Domain	Measure	Human	GPT	Claude
	Effective number of blocks (`effective_types`)	177 ± 7	124 ± 4 (−30%)	165 ± 4 (−7%)
	Share taken by the 50 commonest (`cover_50`)	64.1% ± 0.7	70.3% ± 0.6 (+6.2 pts)	65.2% ± 0.4 (+1.1 pts)
WritingPrompts	Distinct blocks used (`types`)	11,118 ± 285	4,699 ± 138 (−58%)	6,884 ± 155 (−38%)
	Blocks used 5+ times (`types_min5`)	1,583 ± 42	1,091 ± 35 (−31%)	1,371 ± 44 (−13%)
	Effective number of blocks (`effective_types`)	190 ± 5	100 ± 2 (−47%)	136 ± 2 (−28%)
	Share taken by the 50 commonest (`cover_50`)	66.2% ± 0.4	76.1% ± 0.3 (+9.9 pts)	70.4% ± 0.3 (+4.2 pts)

† Non-significant results (corrected p ≥ 0.05).

Effective number of blocks (`effective_types`): the number of equally common building blocks that would give the same variety as the real, uneven usage.

A raw count of distinct blocks treats a block used once the same as a block used thousands of times. The effective number weights each block by how much it is actually used, so a block used once adds very little and a block used thousands of times adds a lot. For example, usage counts of 50, 30, 10, 5 and 5 are five blocks, but two of them account for 80% of the use, so the effective number is 3.45. In these data human essays use 6,954 different blocks, yet their effective number is 133: their variety is comparable to 133 equally used blocks, because a few common blocks dominate. It is a measure of the toolbox in practice and not its full reach, it is less affected by one-off blocks than the raw count, and it falls when usage concentrates on fewer blocks. Technically it is `exp(H)` , where `H = -sum(p ln p)` is the Shannon entropy of each block's share p of all use (a Hill number of order 1).

Other notes:

- **±** is a margin of error: 1.96 × the standard error, about a 95% range. The dagger sits next to it because it says the difference is within that noise.
- **Percentages** in brackets are each AI value relative to human, so (−35%) means 35% below human. `cover_50` is already a percentage, so its difference is in percentage points (pts).

- **Direction:** a lower value on the first three measures means a narrower repertoire, and a higher `cover_50` means more of the text rests on the commonest blocks, which is also narrower.
- **Significance** uses Benjamini–Hochberg correction across the 24 comparisons in the table.

Table 2. Check: the same number of building blocks, not the same amount of text

Table 1 gives every source the same amount of text. This check instead gives every source the same number of building blocks (264,587 block occurrences in Student essays, 230,608 block occurrences in Reuters news and 222,994 block occurrences in WritingPrompts), and everything else is unchanged. It is needed because a source could look narrower simply by building fewer blocks per word, which equal text would not reveal.

Domain	Measure	Human	GPT	Claude
Student essays	Distinct blocks used (`types`)	6,907 ± 319	4,514 ± 179 (−35%)	6,233 ± 165 (−10%)
	Blocks used 5+ times (`types_min5`)	1,380 ± 43	1,149 ± 37 (−17%)	1,367 ± 43† (−1%)
	Effective number of blocks (`effective_types`)	133 ± 4	113 ± 2 (−15%)	134 ± 2† (+1%)
	Share taken by the 50 commonest (`cover_50`)	70.5% ± 0.5	73.1% ± 0.3 (+2.6 pts)	70.0% ± 0.3† (−0.5 pts)
Reuters news	Distinct blocks used (`types`)	7,196 ± 318	4,147 ± 175 (−42%)	6,002 ± 169 (−17%)
	Blocks used 5+ times (`types_min5`)	1,573 ± 56	1,192 ± 41 (−24%)	1,482 ± 47 (−6%)
	Effective number of blocks (`effective_types`)	177 ± 7	124 ± 4 (−30%)	165 ± 4 (−7%)
	Share taken by the 50 commonest (`cover_50`)	64.1% ± 0.7	70.3% ± 0.6 (+6.2 pts)	65.2% ± 0.4 (+1.1 pts)
WritingPrompts	Distinct blocks used (`types`)	11,247 ± 317	4,586 ± 142 (−59%)	6,884 ± 161 (−39%)

Domain	Measure	Human	GPT	Claude
	Blocks used 5+ times (`types_min5`)	1,600 ± 44	1,065 ± 35 (−33%)	1,371 ± 45 (−14%)
	Effective number of blocks (`effective_types`)	191 ± 5	100 ± 2 (−48%)	136 ± 2 (−28%)
	Share taken by the 50 commonest (`cover_50`)	66.2% ± 0.4	76.1% ± 0.3 (+9.9 pts)	70.4% ± 0.3 (+4.2 pts)

† Non-significant results (corrected p ≥ 0.05).

Other notes:

- **±** is a margin of error: 1.96 × the standard error, about a 95% range. The dagger sits next to it because it says the difference is within that noise.
- **Percentages** in brackets are each AI value relative to human, so (−35%) means 35% below human. `cover_50` is already a percentage, so its difference is in percentage points (pts).
- **Direction:** a lower value on the first three measures means a narrower repertoire, and a higher `cover_50` means more of the text rests on the commonest blocks, which is also narrower.
- **Significance** uses Benjamini–Hochberg correction across the 24 comparisons in the table.
- The measures are defined under Table 1.

Table 3. Check: larger building blocks (depth 3)

Tables 1 and 2 use the smallest building blocks. This check uses larger ones: a phrase, its parts, and the parts of those parts, such as `(VP (VBD) (NP (DT) (NN)))`. It is needed because a source could be narrow in its simplest constructions but not in its more elaborate ones, or the reverse. Larger blocks are far more varied, so almost every one is rare: counts are much larger, few blocks reach five uses, and the share taken by the 50 commonest is much smaller. The sizes are therefore not comparable with Table 1, only the direction of the differences. It uses the same amount of text as Table 1 (413,819 tokens in Student essays, 358,204 tokens in Reuters news and 368,665 tokens in WritingPrompts).

Domain	Measure	Human	GPT	Claude
Student essays	Distinct blocks used (`types`)	37,528 ± 747	28,286 ± 530 (−25%)	36,325 ± 453 (−3%)

Domain	Measure	Human	GPT	Claude
	Blocks used 5+ times (`types_min5`)	3,588 ± 73	3,538 ± 74† (−1%)	3,724 ± 87 (+4%)
	Effective number of blocks (`effective_types`)	2,202 ± 85	1,636 ± 44 (−26%)	2,317 ± 55 (+5%)
	Share taken by the 50 commonest (`cover_50`)	35.1% ± 0.5	37.4% ± 0.3 (+2.2 pts)	34.7% ± 0.3† (−0.4 pts)
Reuters news	Distinct blocks used (`types`)	37,119 ± 1,020	25,895 ± 712 (−30%)	35,091 ± 668 (−5%)
	Blocks used 5+ times (`types_min5`)	3,599 ± 74	3,449 ± 57 (−4%)	3,564 ± 83† (−1%)
	Effective number of blocks (`effective_types`)	3,006 ± 157	1,707 ± 75 (−43%)	2,872 ± 105† (−4%)
	Share taken by the 50 commonest (`cover_50`)	30.7% ± 0.6	35.0% ± 0.5 (+4.3 pts)	31.0% ± 0.4† (+0.3 pts)
WritingPrompts	Distinct blocks used (`types`)	42,902 ± 581	26,798 ± 413 (−38%)	34,393 ± 407 (−20%)
	Blocks used 5+ times (`types_min5`)	3,302 ± 68	3,061 ± 64 (−7%)	3,282 ± 78† (−1%)
	Effective number of blocks (`effective_types`)	3,315 ± 90	1,455 ± 37 (−56%)	2,437 ± 53 (−26%)
	Share taken by the 50 commonest (`cover_50`)	29.6% ± 0.3	39.1% ± 0.3 (+9.4 pts)	32.5% ± 0.3 (+2.8 pts)

† Non-significant results (corrected p ≥ 0.05).

Other notes:

- **±** is a margin of error: 1.96 × the standard error, about a 95% range. The dagger sits next to it because it says the difference is within that noise.
- **Percentages** in brackets are each AI value relative to human, so (−35%) means 35% below human. `cover_50` is already a percentage, so its difference is in percentage points (pts).

- **Direction:** a lower value on the first three measures means a narrower repertoire, and a higher `cover_50` means more of the text rests on the commonest blocks, which is also narrower.
- **Significance** uses Benjamini–Hochberg correction across the 24 comparisons in the table.
- The measures are defined under Table 1.
- Compare directions, not sizes, with Table 1.